

## Appendix: Free-Energy and Active-Inference Formalisms

## Appendix: Free-Energy and Active-Inference Formalisms I

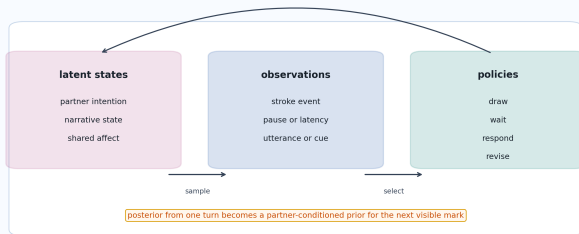
This appendix holds the mathematical and diagnostic primitives used by the main manuscript. The consolidation is deliberate: active inference remains a modeling lens in the main line, while the formal details sit here where readers can inspect assumptions, generators, labels, and evidence boundaries together. The appendix draws on established active-inference and free-energy treatments [friston2010fep;friston2017process;parr2022activeinference;friston2023simpler], discrete-state synthesis work [daco sta2020discrete], and interactional accounts of communication and second-person modeling [vasil2020communication;schilbach2013secondperson;redcay2019secondperson;bolis2024secondperson].

# Event-to-Model Mapping I

[@fig:active\_inference\_mapping] maps DigiPPPiP's theoretical terms to observable design elements. Latent states are partner intentions, narrative states, and shared affect. Observations are strokes, pauses, utterances, and interface events. Policies are drawing, waiting, responding, and revising. The figure is included to prevent active inference from becoming decorative terminology: every model term has to be tied to a protocol event or dropped.

# Event-to-Model Mapping II

Active-inference variables mapped to observable drawing events



CONCEPTUAL

generate\_active\_inference\_mapping | Deterministic project-code figure; not participant outcome data unless stated.

Figure 1: Mapping from DigiPPPiP events to active-inference model components. The figure is generated by `generate_active_inference_mapping()`; read the columns as latent states, observations, and policies, with arrows showing how partner-conditioned beliefs can update through visible marks. It supports the active-inference appendix by tying terms to measurable protocol events. The mapping is conceptual; fitted parameters and failed-baseline checks would be needed for mechanism claims.

# Minimal Free-Energy Primitive I

The project uses a Gaussian point-belief model as a minimal conceptual primitive. Variational free energy is:

$$F(\mu) = \frac{1}{2}\pi_p(\mu - \mu_p)^2 + \frac{1}{2}\pi_l(o - \mu)^2 - \frac{1}{2}\log \pi_p - \frac{1}{2}\log \pi_l$$

{#eq:vfe}

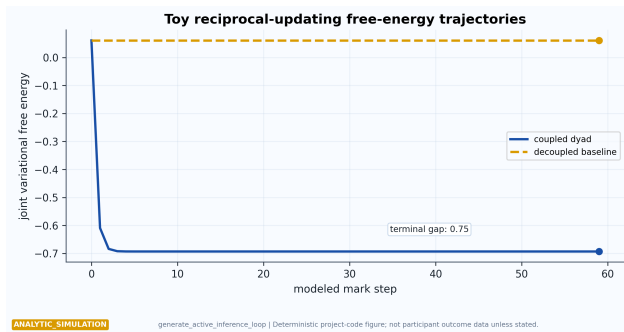
The expression in [eq:vfe] defines  $\mu$  as the current belief,  $\mu_p$  as the prior mean,  $o$  as the canvas observation, and  $\pi_p, \pi_l$  as the prior and likelihood precisions. The analytic posterior mean used by `src/active_inference.py` is:

$$\mu^* = \frac{\pi_p \mu_p + \pi_l o}{\pi_p + \pi_l}$$

{#eq:posterior}

# Minimal Free-Energy Primitive II

In a coupled dyad, each partner's previous posterior becomes part of the other's next prior. [fig:active\_inference\_loop] contrasts that coupled case with a decoupled baseline. The terminal free-energy values from the deterministic conceptual run are -0.693147 and 0.061153, respectively; the absolute reduction is 0.7543. These are model outputs, not human-subject measurements.



## Minimal Free-Energy Primitive III

Figure 2: Coupled and decoupled conceptual free-energy trajectories generated by `generate_active_inference_loop()` from `simulate_d`  
`yadic_session()`. Read the solid and dashed lines as toy-model baselines for reciprocal versus non-reciprocal prior updating across mark steps. The plot supports the modeling argument that coupling can be formalized, not that couples improve. Empirical validation would require observed behavior and model-comparison evidence.

# Hyperscanning and Network Geometry I



## Hyperscanning and Network Geometry II

Geometric hyperscanning is a candidate measurement vocabulary for empirical studies, not a shortcut to interpreting synchrony. Near-infrared spectroscopy (NIRS)-based cooperation studies and hyperscanning reviews motivate dyadic measurement [ @cui2012nirs ; @czeszumski2020hyperscanning; @czeszumski2022cooperative; @nam2020hyperscanningreview], yet shared stimulus timing, motion artifacts, physiology, task structure, preprocessing choices, and analytic windows can all inflate apparent coupling. Methodological cautions in fNIRS and hyperscanning warn against interpreting cross-brain coherence as direct evidence of shared minds, empathy, or relationship improvement [ @tachtsidis2016fnirs; @hamilton2021hyperscanning; @zimmermann2024methodological]. Translational hyperscanning arguments are valuable only when they remain close to applied contexts and family or dyadic care questions rather than treating synchrony as a generic marker of social quality [ @provenzi2022translational]. [ @fig:network\_analysis\_pipeline] therefore separates acquisition, cleaning, windowing, synchrony estimation, graph construction, and curvature analysis before interpretation.

# Narrative Information and Aesthetic Discovery I

Narrative also fits the active-inference vocabulary. Narratives guide event segmentation, identity coherence, and policy selection [bouizegarene2024narrative;veisserie2020thinking]. In DigiPPPiP, a mark sequence can be treated as a symbolic trajectory with Shannon entropy and surprisal [shannon1948mathematical]:

$$H(X) = - \sum_i p(x_i) \log_2 p(x_i)$$

{#eq:shannon}

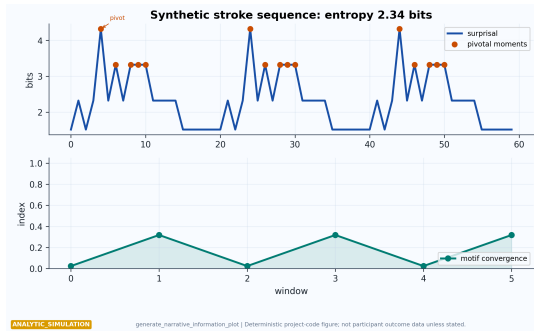
The expected-information-gain arc used for aesthetic co-discovery is:

$$\text{EIG}(t) = c t e^{-\lambda t}$$

{#eq:epistemic\_arc}

# Narrative Information and Aesthetic Discovery II

[@fig:narrative\_information] and [@fig:epistemic\_arc] translate those commitments into inspectable conceptual plots. The configured alphabet supports a maximum entropy of 2.585 bits, and the epistemic arc peaks at model step 2.



# Narrative Information and Aesthetic Discovery III

Figure 6: Narrative-information diagnostics for a conceptual stroke sequence. The figure is generated by `generate_narrative_information_plot()` from `src/narrative.py`; read the upper panel as surprisal with pivotal-moment markers and the lower panel as convergence toward a repeated motif. It supports the narrative-information appendix by making the entropy primitive inspectable. The sequence is synthetic, and participant claims would require coded stroke alphabets and inter-rater checks.

# Narrative Information and Aesthetic Discovery IV

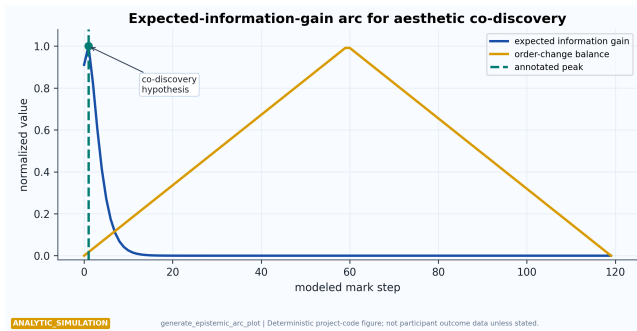


Figure 7: Expected-information-gain and order-change balance in a conceptual DigiPPPiP session. The figure is generated by `generate_epistemic_arc_plot()` from `src/aesthetics.py`; read the normalized curves as an illustrative relation between curiosity, order, change, and the marked “aha” peak. It supports the appendix’s modeling vocabulary. The plot is conceptual, and validation would require event-aligned experience reports or behavioral measures.

## Narrative Information and Aesthetic Discovery V

Together, these equations function as a falsifiable grammar for future studies. They succeed only if they help specify what should be measured, what comparison would matter, and what result would count against the interpretation. A free-energy curve that improves only because of parameter choices, a synchrony measure that disappears after artifact controls, or an entropy score unrelated to participant reports would all narrow the framework rather than embarrass it.